

TED UNIVERSITY

Faculty of Engineering

Department of Computer Engineering

SENIOR DESIGN PROJECT PROPOSAL

Project Title: AlerTED - Offline AI-Powered Emergency Medical Triage and Decision Support System

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Web Site: https://berkealbayrak.github.io/alerted/

AlerTED is an offline, AI-powered mobile decision-support application designed for emergency situations such as disasters, clashes or operations in remote areas where internet connectivity is limited or unavailable. It aims to assist individuals, particularly those without formal medical training, in providing decision support during basic triage assessments and initial medical response process. The project focuses on evaluating both key indicators of a casualty's vital status and observable signs of traumatic injury. Through a series of clear step by step questions the system gathers essential information about the casualty's condition; simultaneously, it uses AI-based image processing- analyzing footage from the mobile devices camera to identify and provide further details on visible traumatic findings such as open wounds, burns or active bleeding. The visual data and the information provided by the user are processed by a decision-support mechanism built upon established triage protocols, helping to determine initial medical response process.

AlerTED will enable the user to access the appropriate basic first-aid procedures that start with the person most in need and will provide the necessary first-aid steps. The application will be developed not for the purpose of making definitive medical diagnoses or replacing professional healthcare services, but rather to provide decision support to individuals in situations where professional assistance is delayed or communication is interrupted. Both the image analysis and the decision-support processes are designed to function entirely on the mobile device without requiring an internet connection. AlerTED is being developed as a research project with its performance to be evaluated under controlled conditions.